The Exponent Rules

Exponent Rules Learned in Pre-Algebra

I) $x^a \cdot x^b = x^{a+b}$ Example: $x^2 \cdot x^5 = x^{2+5} = x^7$

II) $\frac{x^a}{x^b} = x^{a-b}$, $x \neq 0$ Example: $\frac{x^9}{x^3} = x^{9-3} = x^6$

New Exponent Rules

III) $(xy)^a = x^a \cdot y^a$ Example: $(2m)^4 = 2^4 \cdot m^4$

IV) $\left(\frac{x}{y}\right)^a = \frac{x^a}{y^a}$, $y \neq 0$ Example: $\left(\frac{n}{3}\right)^7 = \frac{n^7}{3^7}$

V) $(x^a)^b = x^{ab}$ Example: $(x^{10})^3 = x^{10 \cdot 3} = x^{30}$

Strange New Exponent Rules

VI) $x^1 = x$ Example: $3^1 = 3$

VII) $x^0 = 1$, $x \neq 0$ Example: $5^0 = 1$

VIII) $x^{-a} = \frac{1}{x^a}$, $x \neq 0$ Example: $x^{-2} = \frac{1}{x^2}$

IX) $x^{\frac{1}{a}} = \sqrt[a]{x}$ Example: $x^{\frac{1}{3}} = \sqrt[3]{x}$

X) $x^{\frac{b}{a}} = (\sqrt[a]{x})^b$ or $\sqrt[a]{x^b}$ Example: $x^{\frac{5}{7}} = (\sqrt[7]{x})^5$ or $\sqrt[7]{x^5}$

Name
Date
Course

Using the Chain Rule with the Product Rule or the Quotient Rule

Differentiation Formulas

Constant Rule i) $\frac{d}{dx}(c)=0$ "c" is a constant.

Power Rule ii) $\frac{d}{dx}(x^n)=nx^{n-1}$ "n" is a constant.

Constant Multiple Rule iii) $\frac{d}{dx}[cf(x)]=cf'(x)$ "c" is a constant.

Sum Rule iv) $\frac{d}{dx}[f(x)+g(x)]=f'(x)+g'(x)$

Difference Rule v) $\frac{d}{dx}[f(x)-g(x)]=f'(x)-g'(x)$

Product Rule vi) $\frac{d}{dx}[f(x)g(x)]=f'(x)g(x)+f(x)g'(x)$

Quotient Rule vii) $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right]=\frac{f'(x)g(x)-f(x)g'(x)}{[g(x)]^2}$

The Chain Rule viii) If $h(x)=f(g(x))$, then

$$\boxed{h'(x)=f'(g(x))g'(x)}$$

Three Steps For the Chain Rule

1) Find the derivative of the outside function while
2) leaving the inside function unchanged.
3) Multiply the resulting expression by the derivative of the inside function.

Trigonometric Differentiation Formulas

I) $\frac{d}{dx}(\sin x)=\cos x$

II) $\frac{d}{dx}(\cos x)=-\sin x$

III) $\frac{d}{dx}(\csc x)=-\csc x\cot x$

IV) $\frac{d}{dx}(\sec x)=\sec x\tan x$

V) $\frac{d}{dx}(\tan x)=\sec^2 x$

VI) $\frac{d}{dx}(\cot x)=-\csc^2 x$

Using the Chain Rule with the Product Rule

Find $f'(x)$.

① $f(x) = x^3\cos(5x^4)$

$$f'(x) = \left[\frac{d}{dx}(x^3)\right]\cos(5x^4) + x^3\frac{d}{dx}\left(\cos(5x^4)\right)$$

$$= 3x^2\cos(5x^4) + x^3\left[-\sin(5x^4)\right]\frac{d}{dx}(5x^4)$$

$$= 3x^2\cos(5x^4) - x^3\left[\sin(5x^4)\right]20x^3$$

$$= \boxed{3x^2\cos(5x^4) - 20x^6\sin(5x^4)}$$

② $f(x) = (3x^2+2)^4x^2$

$$f'(x) = \left[\frac{d}{dx}\left((3x^2+2)^4\right)\right]x^2 + (3x^2+2)^4\frac{d}{dx}(x^2)$$

$$= 4(3x^2+2)^{4-1}\left[\frac{d}{dx}(3x^2+2)\right]x^2 + (3x^2+2)^4(2x)$$

$$= 4(3x^2+2)^3(6x)\,x^2 + 2x(3x^2+2)^4$$

$$= \boxed{24x^3(3x^2+2)^3 + 2x(3x^2+2)^4}$$

③ $f(x) = x^4\sqrt[3]{2+4x^2+x^3}$

$$f'(x) = \left[\frac{d}{dx}(x^4)\right]\sqrt[3]{2+4x^2+x^3} + x^4\frac{d}{dx}\left(\sqrt[3]{2+4x^2+x^3}\right)$$

$$f'(x) = 4x^3\sqrt[3]{2+4x^2+x^3} + x^4\frac{d}{dx}\left((2+4x^2+x^3)^{\frac{1}{3}}\right)$$

$$f'(x) = 4x^3\sqrt[3]{2+4x^2+x^3} + x^4\,\frac{1}{3}(2+4x^2+x^3)^{\frac{1}{3}-1}\frac{d}{dx}(2+4x^2+x^3)$$

$$f'(x) = 4x^3\sqrt[3]{2+4x^2+x^3} + \frac{1}{3}x^4(2+4x^2+x^3)^{-\frac{2}{3}}(8x+3x^2)$$

$$f'(x) = 4x^3\sqrt[3]{2+4x^2+x^3} + \frac{1}{3}x^4\,\frac{1}{\sqrt[3]{(2+4x^2+x^3)^2}}(8x+3x^2)$$

$$f'(x) = \boxed{4x^3\sqrt[3]{2+4x^2+x^3} + \frac{8x^5+3x^6}{3\sqrt[3]{(2+4x^2+x^3)^2}}}$$

④ $f(x)=x^5\sin(-2x^2)$

$$f'(x)=\left[\frac{d}{dx}(x^5)\right]\sin(-2x^2)+x^5\frac{d}{dx}(\sin(-2x^2))$$

$$f'(x)=5x^4\sin(-2x^2)+x^5[\cos(-2x^2)]\frac{d}{dx}(-2x^2)$$

$$f'(x)=5x^4\sin(-2x^2)+x^5[\cos(-2x^2)](-4x)$$

$$f'(x)=\boxed{5x^4\sin(-2x^2)-4x^6\cos(-2x^2)}$$

⑤ $f(x)=(x^3+7x^2+x-8)^3x^6$

$$f'(x)=\left[\frac{d}{dx}((x^3+7x^2+x-8)^3)\right]x^6+(x^3+7x^2+x-8)^3\frac{d}{dx}(x^6)$$

$$f'(x)=\left[3(x^3+7x^2+x-8)^{3-1}\frac{d}{dx}(x^3+7x^2+x-8)\right]x^6+(x^3+7x^2+x-8)^3 6x^5$$

$$f'(x)=\boxed{3x^6(x^3+7x^2+x-8)^2(3x^2+14x+1)+6x^5(x^3+7x^2+x-8)^3}$$

⑥ $f(x)=x^7\sqrt[4]{10x-7}$

$$f'(x)=\left[\frac{d}{dx}(x^7)\right]\sqrt[4]{10x-7}+x^7\frac{d}{dx}(\sqrt[4]{10x-7})$$

$$f'(x)=7x^6\sqrt[4]{10x-7}+x^7\frac{d}{dx}\left[(10x-7)^{\frac{1}{4}}\right]$$

$$f'(x)=7x^6\sqrt[4]{10x-7}+x^7\frac{1}{4}(10x-7)^{\frac{1}{4}-1}\frac{d}{dx}(10x-7)$$

$$f'(x)=7x^6\sqrt[4]{10x-7}+\frac{1}{4}x^7(10x-7)^{-3/4}(10)$$

$$f'(x)=\boxed{7x^6\sqrt[4]{10x-7}+\frac{5x^7}{2\sqrt[4]{(10x-7)^3}}}$$

Using the Chain Rule Twice

Differentiate.

⑦ $f(x) = (-x+1)^5 \tan\sqrt{x}$

$$f'(x) = \left[\frac{d}{dx}(-x+1)^5\right]\tan\sqrt{x} + (-x+1)^5\frac{d}{dx}(\tan\sqrt{x})$$

$$f'(x) = \left[5(-x+1)^{5-1}\frac{d}{dx}(-x+1)\right]\tan\sqrt{x} + (-x+1)^5\frac{d}{dx}(\tan(x^{\frac{1}{2}}))$$

$$f'(x) = 5(-x+1)^4(-1)\tan\sqrt{x} + (-x+1)^5\sec^2(x^{\frac{1}{2}})\frac{d}{dx}(x^{\frac{1}{2}})$$

$$f'(x) = -5(-x+1)^4\tan\sqrt{x} + (-x+1)^5\sec^2(\sqrt{x})\,\frac{1}{2}x^{\frac{1}{2}-1}$$

$$f'(x) = \boxed{-5(-x+1)^4\tan\sqrt{x} + \frac{(-x+1)^5\sec^2\sqrt{x}}{2\sqrt{x}}}$$

⑧ $f(x) = (8-2x+x^8)^2\cot(x^9-2x)$

$$f'(x) = \left[\frac{d}{dx}\left((8-2x+x^8)^2\right)\right]\cot(x^9-2x) + (8-2x+x^8)^2\frac{d}{dx}(\cot(x^9-2x))$$

$$f'(x) = \left[2(8-2x+x^8)^{2-1}\frac{d}{dx}(8-2x+x^8)\right]\cot(x^9-2x) + (8-2x+x^8)^2(-\csc^2(x^9-2x))\frac{d}{dx}(x^9-2x)$$

$$f'(x) = 2(8-2x+x^8)(-2+8x^7)\cot(x^9-2x) - (8-2x+x^8)^2\left[\csc^2(x^9-2x)\right](9x^8-2)$$

$$f'(x) = \boxed{2(8-2x+x^8)(-2+8x^7)\cot(x^9-2x) - (8-2x+x^8)^2(9x^8-2)\csc^2(x^9-2x)}$$

⑨ $f(x) = (3x-1)^3\sec(x^2)$

$$f'(x) = \left[\frac{d}{dx}((3x-1)^3)\right]\sec(x^2) + (3x-1)^3\frac{d}{dx}(\sec(x^2))$$

$$f'(x) = \left[3(3x-1)^{3-1}\frac{d}{dx}(3x-1)\right]\sec(x^2) + (3x-1)^3\sec(x^2)\tan(x^2)\frac{d}{dx}(x^2)$$

$$f'(x) = 3(3x-1)^2(3)\sec(x^2) + (3x-1)^3\sec(x^2)\left[\tan(x^2)\right]2x$$

$$f'(x) = \boxed{9(3x-1)^2\sec(x^2) + 2x(3x-1)^3\sec(x^2)\tan(x^2)}$$

(10) $f(x) = \sqrt{x^2-9}\,\csc(x^3)$

$$f'(x) = \left[\frac{d}{dx}\left(\sqrt{x^2-9}\right)\right]\csc(x^3) + \sqrt{x^2-9}\,\frac{d}{dx}\left(\csc(x^3)\right)$$

$$f'(x) = \left[\frac{d}{dx}\left((x^2-9)^{\frac{1}{2}}\right)\right]\csc(x^3) + \sqrt{x^2-9}\left(-\csc(x^3)\cot(x^3)\frac{d}{dx}(x^3)\right)$$

$$f'(x) = \left[\frac{1}{2}(x^2-9)^{\frac{1}{2}-1}\frac{d}{dx}(x^2-9)\right]\csc(x^3) - \sqrt{x^2-9}\,\csc(x^3)\left[\cot(x^3)\right]3x^2$$

$$f'(x) = \frac{1}{2}(x^2-9)^{-\frac{1}{2}}(2x)\csc(x^3) - 3x^2\sqrt{x^2-9}\,\csc(x^3)\cot(x^3)$$

$$f'(x) = \frac{1}{2}\,\frac{1}{(x^2-9)^{1/2}}(2x)\csc(x^3) - 3x^2\sqrt{x^2-9}\,\csc(x^3)\cot(x^3)$$

$$= \frac{2x\csc(x^3)}{2\sqrt{x^2-9}} - 3x^2\sqrt{x^2-9}\,\csc(x^3)\cot(x^3)$$

$$= \boxed{\frac{x\csc(x^3)}{\sqrt{x^2-9}} - 3x^2\sqrt{x^2-9}\,\csc(x^3)\cot(x^3)}$$

Using the Chain Rule with the Quotient Rule

Find $\frac{dy}{dx}$.

(11) $y = \frac{x}{\sin(6x^5)}$

$$\frac{dy}{dx} = \frac{\left[\frac{d}{dx}(x)\right]\sin(6x^5) - x\,\frac{d}{dx}\left(\sin(6x^5)\right)}{\left(\sin(6x^5)\right)^2}$$

$$\frac{dy}{dx} = \frac{(1)\sin(6x^5) - x\cos(6x^5)\,\frac{d}{dx}(6x^5)}{\sin^2(6x^5)}$$

$$\frac{dy}{dx} = \frac{\sin(6x^5) - x\left[\cos(6x^5)\right](30x^4)}{\sin^2(6x^5)}$$

$$\frac{dy}{dx} = \boxed{\frac{\sin(6x^5) - 30x^5\cos(6x^5)}{\sin^2(6x^5)}}$$

(12) $y = \frac{(3x+5)^3}{x^4+10}$

$$\frac{dy}{dx} = \frac{\left[\frac{d}{dx}((3x+5)^3)\right](x^4+10) - (3x+5)^3 \frac{d}{dx}(x^4+10)}{(x^4+10)^2}$$

$$\frac{dy}{dx} = \frac{\left[3(3x+5)^{3-1}\frac{d}{dx}(3x+5)\right](x^4+10) - (3x+5)^3(4x^3)}{(x^4+10)^2}$$

$$\frac{dy}{dx} = \frac{3(3x+5)^2(3)(x^4+10) - 4x^3(3x+5)^3}{(x^4+10)^2}$$

$$\frac{dy}{dx} = \boxed{\frac{9(3x+5)^2(x^4+10) - 4x^3(3x+5)^3}{(x^4+10)^2}}$$

$$\frac{dy}{dx} = \frac{(3x+5)^2\left[9(x^4+10) - 4x^3(3x+5)\right]}{(x^4+10)^2}$$

$$\frac{dy}{dx} = \boxed{-\frac{(3x+5)^2(3x^4+20x^3-90)}{(x^4+10)^2}}$$

(13) $y = \frac{\cos(x^2-1)}{4x+7}$

$$\frac{dy}{dx} = \frac{\left[\frac{d}{dx}(\cos(x^2-1))\right](4x+7) - \cos(x^2-1)\frac{d}{dx}(4x+7)}{(4x+7)^2}$$

$$\frac{dy}{dx} = \frac{\left[-\sin(x^2-1)\frac{d}{dx}(x^2-1)\right](4x+7) - \left[\cos(x^2-1)\right](4)}{(4x+7)^2}$$

$$\frac{dy}{dx} = \frac{\left[-\sin(x^2-1)\right](2x)(4x+7) - 4\cos(x^2-1)}{(4x+7)^2}$$

$$\frac{dy}{dx} = \boxed{-\frac{(8x^2+14x)\sin(x^2-1) + 4\cos(x^2-1)}{(4x+7)^2}}$$

(14) $y = \frac{5x^3-2x}{(x-4)^4}$

$$\frac{dy}{dx} = \frac{\left[\frac{d}{dx}(5x^3-2x)\right](x-4)^4 - (5x^3-2x)\frac{d}{dx}((x-4)^4)}{\left[(x-4)^4\right]^2}$$

$$\frac{dy}{dx} = \frac{(15x^2-2)(x-4)^4 - (5x^3-2x)\,4\,(x-4)^{4-1}\frac{d}{dx}(x-4)}{(x-4)^8}$$

$$\frac{dy}{dx} = \frac{(15x^2-2)(x-4)^4 - 4(5x^3-2x)(x-4)^3(1)}{(x-4)^8}$$

$$\frac{dy}{dx} = \frac{(x-4)^3\left[(15x^2-2)(x-4) - 4(5x^3-2x)\right]}{(x-4)^3(x-4)^5}$$

$$\frac{dy}{dx} = \frac{(15x^2-2)(x-4) - 20x^3+8x}{(x-4)^5}$$

$$\frac{dy}{dx} = \frac{15x^3-60x^2-2x+8-20x^3+8x}{(x-4)^5}$$

$$\frac{dy}{dx} = \boxed{\frac{-5x^3-60x^2+6x+8}{(x-4)^5}}$$

Find y'.

(15) $y = \frac{4}{(x^3-4)^5}$

$$y' = \frac{\left[\frac{d}{dx}(4)\right](x^3-4)^5 - 4\frac{d}{dx}((x^3-4)^5)}{\left[(x^3-4)^5\right]^2}$$

$$y' = \frac{0\,(x^3-4)^5 - 4\left[5(x^3-4)^{5-1}\frac{d}{dx}(x^3-4)\right.}{(x^3-4)^{10}}$$

$$y' = \frac{0 - 20(x^3-4)^4(3x^2)}{(x^3-4)^{10}}$$

$$y' = \frac{-60x^2(x^3-4)^4}{(x^3-4)^4(x^3-4)^6}$$

$$y' = \boxed{-\frac{60x^2}{(x^3-4)^6}}$$

16) $y = -\frac{2}{\sqrt{6x-1}}$

$$y = \frac{-2}{\sqrt{6x-1}}$$

$$y' = \frac{\left[\frac{d}{dx}(-2)\right]\sqrt{6x-1} - (-2)\frac{d}{dx}\left(\sqrt{6x-1}\right)}{\left(\sqrt{6x-1}\right)^2}$$

$$y' = \frac{(0)\sqrt{6x-1} + 2\frac{d}{dx}\left((6x-1)^{1/2}\right)}{6x-1}$$

$$y' = \frac{0 + 2\left(\frac{1}{2}(6x-1)^{\frac{1}{2}-1}\right)\frac{d}{dx}(6x-1)}{6x-1}$$

$$y' = \frac{(6x-1)^{-\frac{1}{2}}(6)}{(6x-1)^1}$$

$$y' = 6(6x-1)^{-\frac{1}{2}-1}$$

$$y' = 6(6x-1)^{-\frac{3}{2}}$$

$$y' = \boxed{\frac{6}{\sqrt{(6x-1)^3}}}$$

17) $y = \frac{-1}{\csc^2 x + 10}$

$$y' = \frac{\left[\frac{d}{dx}(-1)\right](\csc^2 x + 10) - (-1)\frac{d}{dx}(\csc^2 x + 10)}{(\csc^2 x + 10)^2}$$

$$y' = \frac{(0)(\csc^2 x + 10) + (1)\frac{d}{dx}\left[(\csc x)^2 + 10\right]}{(\csc^2 x + 10)^2}$$

$$y' = \frac{0 + 2(\csc x)^{2-1}\frac{d}{dx}(\csc x)}{(\csc^2 x + 10)^2}$$

$$y' = \frac{2(\csc x)^1(-\csc x \cot x)}{(\csc^2 x + 10)^2}$$

$$y' = \boxed{-\frac{2\csc^2 x \cot x}{(\csc^2 x + 10)^2}}$$

(18) $y = \dfrac{8}{4-\sec^3 x}$

$$y' = \frac{\boxed{\frac{d}{dx}(8)}(4-\sec^3 x) - 8\,\frac{d}{dx}(4-\sec^3 x)}{(4-\sec^3 x)^2}$$

$$y' = \frac{0\,(4-\sec^3 x) - 8\,\frac{d}{dx}(4-(\sec x)^3)}{(4-\sec^3 x)^2}$$

$$y' = \frac{0-8\left(0-3(\sec x)^{3-1}\frac{d}{dx}(\sec x)\right)}{(4-\sec^3 x)^2}$$

$$y' = \frac{-8(-3(\sec x)^2 \sec x \tan x}{(4-\sec^3 x)^2}$$

$$y' = \boxed{\frac{24\sec^3 x \tan x}{(4-\sec^3 x)^2}}$$

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Using the Chain Rule Twice

* Differentiate.

(19) $y = \dfrac{(3-7x)^2}{(x^2+4)^3}$

$$\frac{dy}{dx} = \frac{\left[\frac{d}{dx}\left((3-7x)^2\right)\right](x^2+4)^3 - (3-7x)^2\frac{d}{dx}\left[(x^2+4)^3\right]}{\left[(x^2+4)^3\right]^2}$$

$$\frac{dy}{dx} = \frac{\left[2(3-7x)^{2-1}\frac{d}{dx}(3-7x)\right](x^2+4)^3 - (3-7x)^2\, 3(x^2+4)^{3-1}\frac{d}{dx}(x^2+4)}{(x^2+4)^6}$$

$$\frac{dy}{dx} = \frac{2(3-7x)^1(-7)(x^2+4)^3 - 3(3-7x)^2(x^2+4)^2(2x)}{(x^2+4)^6}$$

$$\frac{dy}{dx} = \frac{-14(3-7x)(x^2+4)^3 - 6x(3-7x)^2(x^2+4)^2}{(x^2+4)^6}$$

$$\frac{dy}{dx} = \frac{(x^2+4)^2\left[-14(3-7x)(x^2+4) - 6x(3-7x)^2\right]}{(x^2+4)^2(x^2+4)^4}$$

$$\frac{dy}{dx} = \frac{(x^2+4)^2\left[-14(3-7x)(x^2+4) - 6x(3-7x)^2\right]}{(x^2+4)^2(x^2+4)^4}$$

$$\frac{dy}{dx} = \boxed{\frac{-14(3-7x)(x^2+4) - 6x(3-7x)^2}{(x^2+4)^4}}$$

(20) $y = \dfrac{(3x^3+2)^4}{(6x-1)^5}$

$$y' = \frac{\left[\frac{d}{dx}\left((3x^3+2)^4\right)\right](6x-1)^5 - (3x^3+2)^4 \frac{d}{dx}\left[(6x-1)^5\right]}{\left[(6x-1)^5\right]^2}$$

$$y' = \frac{\left[4(3x^3+2)^{4-1}\frac{d}{dx}(3x^3+2)\right](6x-1)^5 - (3x^3+2)^4\, 5(6x-1)^{5-1}\frac{d}{dx}(6x-1)}{(6x-1)^{10}}$$

$$y' = \frac{4(3x^3+2)^3(9x^2)(6x-1)^5 - 5(3x^3+2)^4(6x-1)^4(6)}{(6x-1)^{10}}$$

$$y' = \frac{36x^2(3x^3+2)^3(6x-1)^5 - 30(3x^3+2)^4(6x-1)^4}{(6x-1)^{10}}$$

$$y' = \frac{(6x-1)^4\left[36x^2(3x^3+2)^3(6x-1) - 30(3x^3+2)^4\right]}{(6x-1)^4(6x-1)^6}$$

$$y' = \frac{(6x-1)^4\left[36x^2(3x^3+2)^3(6x-1) - 30(3x^3+2)^4\right]}{(6x-1)^4(6x-1)^6}$$

$$y' = \boxed{\frac{(216x^3-36x^2)(3x^3+2)^3 - 30(3x^3+2)^4}{(6x-1)^6}}$$